Chapter - 1

1. Define Society. Explain briefly about the types/elements of society in the sociological point of view. (Marks: 4, 10, 5)

Definition of Society (4 Marks)

Society is fundamentally derived from the Latin word *socks*, which translates to companionship or friendship. Sociologically, a society is defined as a group of individuals who are related to each other, united by certain relationships, modes of behavior, geographical territory, and shared cultural expectations. A society possesses a population, occupies a specific territory, shares the same political authority, and maintains a common culture along with a sense of membership and commitment to the collective group. The essence of society reflects the complex interaction and network of social relationships among this group of people.

Types/Elements of Society in the Sociological Point of View (5 Marks)

The sources identify several essential criteria or elements that characterize a society:

1. **Plurality:** A society must be composed of diverse populations, including individuals of all ages, sexes, and various economic statuses.
2. **Stability:** Society has a stable structure in which social life is mainly organized through the division of labor.
3. **Interdependence:** The members of a society are reliant upon each other for survival and to satisfy their needs and aims, establishing the value of each member. For example, a child requires care from its mother until it can feed itself.
4. **Cooperation:** Cooperation is necessary for the survival of human society, ensuring a sense of confidence among members to receive help from others.
5. **Likeness and Differences:** Society requires both likeness (e.g., common needs, works, aims, values) and differences (e.g., variations in age, sex, opinions, ability, and intellectuality) among its members.

Broadly, human societies globally can be categorized into two major types based on their cultural values and composition: Western Societies and Eastern Societies. Western societies typically value achievement, efficiency, progress, and individualism, whereas Eastern societies prioritize achieving high morality, the power of truth, and success in religious activities over materialistic objects.

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2. Define social change. What are the major factors/theories bringing changes in society? Explain the conflict theory of social change and its relevance in Nepal. (Marks: 4, 1+3, 8)

Definition of Social Change (4 Marks)

Social change is defined as the modification in the ways of acting and thinking (social behavior) of the members of society. It involves the progressive transformation of society, leading to modifications in culture, social structure, and behavior over time towards more positive outcomes. Changes are considered universal and should consistently reflect progress.

Major Factors/Theories Bringing Changes in Society (4 Marks)

Major Factors Affecting Social Change:

Social change is influenced by several external and internal factors:

1. **Cultural Changes:** Modifications in cultural systems through invention, discovery, and diffusion introduce new prospects, thus causing social changes.
2. **Demographic Changes:** Shifts in society composition due to factors like high birth rates, low death rates, and migration.
3. **Ideas and Opinions:** The evolution of new ideas and the modification of old ideas greatly impact social change over time.
4. **Social Conflict and Division:** Conflicts and divisions based on characteristics such as class, religion, gender, and caste contribute to social changes.
5. **Environmental Changes:** Alterations in the environment (e.g., climate or natural calamities) directly affect social life and necessitate changes.

Theories of Social Change:

Four major theories attempt to explain social change:

1. **Socio-Cultural Evolution Theory:** Assumes society develops gradually from simple beginnings into more complex forms out of the necessity for a new, higher level of civilization and betterment.
2. **Functionalist Theory:** Views society as a system of interdependent parts that seek stability, where change occurs as an adjustment to restore balance after disruption.
3. **Class Conflict Theory (Conflict Theory):** Assumes conflict between classes is inevitable, leading to the dissolution of the current society and emergence of new societies. Today, conflict theory is applied to almost any social divide, including **race, gender, religion, sexuality, and age.**
4. **Cyclical Theory:** Views each civilization as having a biological life cycle (birth, maturity, old age, death), where its fate is determined by how successfully it copes with internal and external challenges.

Conflict Theory of Social Change and its Relevance in Nepal (8 Marks)

Conflict Theory Explanation:

The Class Conflict Theory, or conflict theory, posits that the fundamental structure of society inherently contains different classes of people. According to this theory, conflicts between these classes are an inevitable outcome over time. This ongoing tension and struggle ultimately results in the dissolution of the existing society and the formation of new social structures, which constitutes social change. Conflict theory suggests that instability and tension, rather than consensus or equilibrium, are the driving forces of social transformation.

Relevance in Nepal:

Guff it up babyyyyyy, The Maoist People's War, The Caste System and Dalit Resistance, The Genz Revolution

3. Discuss the significant contributions of eastern societies to the field of engineering. Provide specific examples of their engineering achievements. (Marks: 4)

1. Ancient China: The Masters of Infrastructure

China's contributions are perhaps the most diverse, ranging from massive civil works to the "Four Great Inventions."

- **The Great Wall:** A masterclass in **structural engineering** and logistics, spanning thousands of miles across varied terrain.
- **The Grand Canal:** The world's longest artificial river. It revolutionized **hydraulic engineering** by using a system of "pound locks" to move boats uphill and downhill.
- **Gunpowder:** Originally an alchemical accident, it led to the birth of **chemical engineering** and ballistics.
- **Suspension Bridges:** China was using iron-chain suspension bridges centuries before they appeared in Europe.

2. Ancient India: Precision and Urban Planning

The Indian subcontinent excelled in civil engineering and high-quality material science.

- **Indus Valley Urban Planning:** Cities like Mohenjo-Daro featured the world's first planned **sewage and drainage systems**, with grid-pattern streets and standardized bricks.
- **The Iron Pillar of Delhi:** A 7-meter-high pillar that has resisted corrosion for over 1,600 years, showing an advanced understanding of extraction and oxidation.

3. Mesopotamia (Modern Iraq): The Birth of Civil Engineering

As the "Cradle of Civilization," Mesopotamia turned nomadic life into organized urban living.

- **The Wheel:** Perhaps the most important **mechanical engineering** invention in human history, originally used for pottery and later for transportation.
- **Irrigation Systems:** They engineered complex canals and dikes to control the unpredictable flooding of the Tigris and Euphrates rivers.

4. The Islamic Golden Age: Automata and Hydraulics

During the Middle Ages, Middle Eastern scholars preserved and expanded upon Greek and Indian knowledge, focusing heavily on machines.

- **Windmills:** The first vertical-axle windmills were developed in Persia (modern Iran) for grinding grain and pumping water.

5. Ancient Egypt: Monumental Engineering

Though often debated as "Middle Eastern" or "African," Egypt's proximity and influence on Eastern trade routes make its contributions vital.

- **The Pyramids:** Specifically the Great Pyramid of Giza, which remained the tallest man-made structure for nearly 4,000 years. It demonstrates near-perfect **geometric alignment** and massive-scale stone transport.

Summary Table: Key Engineering Milestones

Region	**Key Contribution**	**Engineering Field**
China	Grand Canal / Pound Locks	Hydraulic Engineering
India	Planned Sewage / Wootz Steel	Civil / Metallurgical
Mesopotamia	The Wheel / Irrigation	Mechanical / Civil
Persia	Windmills	Energy Engineering

Country	Primary Engineering Strength	Key Innovation
UK	Mechanical & Civil	Steam Engine, Railways
Germany	Automotive & Chemical	Internal Combustion Engine, Haber-Bosch
USA	Industrial & Electronic	Assembly Line, Transistor
France	Structural & Academic	Steel Construction, Thermodynamics
Italy	Electrical & Aerospace	Radio, The Battery

4. Describe the impact of technology (or digital technology/computers) into the society and explain the relationship between human and society/how it helps in economic development. Illustrate the impacts of computer on Nepalese society. (Marks: 3+2, 4, 10)

Impact of Digital Technology/Computers on Society (5 Marks)

The introduction of computers and digital technology has profoundly impacted society in several ways:

1. **Problem Solution:** Computers are essential for solving complex problems efficiently and rapidly, performing tasks impossible for humans, such as in medical diagnosis and quality control.
2. **Productivity and Employment:** The use of computers dramatically increases productivity by completing tasks swiftly. However, this leads to the problem of technological unemployment as computers increasingly overtake human jobs.

3. **Quality of Life:** Digital technology improves the quality of services at lower costs and effort, thus enhancing the overall quality of life by helping people focus on necessary work and eliminate time lost on unnecessary tasks.
4. **Social Interaction:** Computers have transformed the world into a "global village," but the increasing interaction through virtual systems like social media platforms reduces traditional human relationships.

Question: Discuss the relationship between Human and Society. (Marks: 4)

The relationship between humans and society is deeply rooted and interdependent. A human being is a social animal who cannot live in isolation; society provides the framework for human survival, development, and identity.

The following points detail this complex relationship:

1. Meaning and Origin

The term ‘society’ is derived from the Latin word ‘socius’ (meaning companionship or friendship), which highlights that the very foundation of society is human interaction. It is defined as a group of individuals united by specific relations, modes of behavior, and cultural expectations.

2. Mutual Interdependence and Survival

Humans and society are mutually dependent. A person living alone on an isolated island would only need to consider their own needs, but in a group, voluntary actions affect everyone.

- **Survival:** From birth, humans need the cooperation of others for survival (e.g., a child depending on its mother).
- **Need Satisfaction:** Members of a society depend on one another to satisfy their physical, economic, and emotional needs.

3. Mutual Awareness and Likeness

For a society to exist, there must be "reciprocal awareness" among its members. While humans share **likeness** in their basic needs and aims, they also have **differences** in intellect and ability, which allows the society to function through the division of labor.

4. Collaboration and Progress

Society allows humans to work in collaboration, which is far more efficient than individual effort. When humans produced more food than they needed, they began to specialize in other useful tasks, leading to the rise of civilization and technological progress. In modern times, an individual's work is a major part of their identity and their contribution to this societal progress.

5. Social Control and Culture

Every society has its own "way of life," known as **culture**. Society exerts "social control" over individuals through norms, traditions, and customs to ensure stability and orderly behavior. When individual freedom clashes with societal goals, **Ethics** (the study of how to live in a group) provides the norms to resolve these dilemmas.

Impacts of Computer on Nepalese Society (10 Marks)

In Nepal, which is currently undergoing rapid digital transformation, the impact of computer engineering is crucial for modernization:

1. **Digital Governance and E-Services:** Computers and digital systems are fundamental in building and supporting national digital governance and modernization efforts. This includes implementing large-scale systems such as **e-governance platforms**, digital payments, and biometric identity systems like the Nagarik App and National ID.
2. **Financial Inclusion:** Computer engineers contribute directly to inclusive economic development by designing **mobile banking apps** that significantly increase financial access in remote Nepalese villages.
3. **ICT Sector Development:** Institutions like the Computer Association of Nepal (CAN) Federation promote the development of the overall Information and Communication Technology (ICT) sector in Nepal.
4. **Security and Ethical Challenges:** Due to the reliance on sensitive systems, computer engineering introduces high risks of misuse. Ethical responsibilities for engineers in Nepal center around **protecting confidential citizen data**, preventing cybercrime, and ensuring fair digital access in various ICT projects across health, banking, and agriculture.
5. **Professional Accountability:** The Nepal Engineering Council (NEC) ensures that computer engineers adhere to ethical codes in complex areas where misuse of AI, data systems, or software could cause massive harm to the public.

Question: Explain how individual freedom balances societal goals (or why the conflict exists between an individual's freedom and the social goal).

In the study of professional practice and ethics, the relationship between an individual and the society they live in is a fundamental topic. While individuals seek freedom to act according to their own interests, society has collective goals that often require rules and limitations.

1. Why the Conflict Exists

The conflict between individual freedom and societal goals arises primarily because humans do not live in isolation.

- **Interdependence:** If a person lived alone on an isolated island, they would only need to consider themselves. However, when individuals "band together" in a society, their voluntary actions inevitably affect the group.

- **Contrast of Interests:** Conflict occurs when an individual's personal interests (such as maximizing personal profit or total autonomy) contrast with social interests (such as public safety or equality). This creates a "dilemma," where it becomes confusing to determine what is right or wrong.
- **Social Differences:** Societies are composed of people with different interests, opinions, and intellectual levels, which naturally leads to clashing goals.

2. How the Balance is Achieved (The Role of Ethics)

Ethics is the primary tool used to balance these two competing forces. It is often defined as the study of **"how to live in a group"**.

- **Establishing Norms:** Ethics provides the norms or standards for how an individual should behave when their personal desires clash with what is fair or good for the public.
- **Transition to Written Codes:** To maintain morality and achieve societal goals, many communities have moved from unwritten "tribal customs" to formal, written **codes of behavior** or laws.
- **Methods of Analysis:** When an individual faces a dilemma between their freedom and a social goal, they can use three types of analysis to find a balance:
    1. **Economic Analysis:** Looking at the financial costs and benefits.
    2. **Legal Analysis:** Checking what the law of the land requires.
    3. **Philosophical Analysis:** Evaluating the moral "rightness" of the action.

Question 6: Why is society necessary for engineers? What are the roles that an engineer can play in society/Social Change? (Marks: 1+3, 4, 5)

Part 1: Why Society is Necessary for Engineers

For an engineer, society is not just a place to live; it is the very foundation and reason for their profession. The necessity of society for engineers can be explained through the following points:

- **Defining Purpose:** Engineering is defined as the application of scientific knowledge to use materials and forces of nature **for the benefit of mankind**. Without a society (mankind) to serve, the profession would have no objective or purpose.
- **Identity and Contribution:** In modern societies, work is a major part of an individual's identity. Society provides a platform where an engineer's technical skills become a meaningful contribution to progress.
- **Mutual Interdependence:** Humans are social animals who depend on each other for survival and need satisfaction. Engineers need the society to provide the resources, labor, and demand for the infrastructures they design (like bridges, networks, or hospitals).
- **Mission and Motivation:** Society provides the "Mission" for engineers. It presents the problems (like energy shortages or communication gaps) that engineers are motivated to solve economically and safely for the larger public.

- **Trust and Ethics:** A profession exists because society places trust in it. Society provides the ethical and legal frameworks (like building codes and the NEC Act) that guide engineers to practice responsibly.

Part 2: Roles of an Engineer in the Society

Engineers play several critical roles in the developmental activities of a nation. These roles ensure that technology and infrastructure are developed systematically:

Role	Brief Description
Creating Vision	Imagining useful products or systems by utilizing nature's properties for the benefit of people.
Preparing Mission	Planning and preparing to produce goods economically so they are safe, healthy, and accessible to a large number of people.
Execution	Directly implementing or assigning technical tasks that require specialized engineering knowledge.
Monitoring & Evaluation	Supervising work to ensure accuracy, quality, and that projects are completed on time and within budget.
Training & Mentoring	Training new engineers practically and technically to build the next generation of professionals.
Upgrading the Profession	Innovating and creating new facilities while maintaining the dignity and ethical values of the engineering profession.

7. What types of technology/ies Nepal should focus on to develop rural infrastructure rapidly? Be specific to your area of study/program of study. (Marks: 4)

Assuming the program of study is focused on **Computer/Electronics Engineering** (based on the primary context of the provided sources), the critical area for developing rapid rural infrastructure in Nepal is **Information and Communication Technology (ICT)** and related digital systems.

Nepal should focus on the following technologies to bridge the rural development gap:

- **Telecommunications Expansion (6G/5G/Fiber Optics):** Rapid expansion of communication networks is vital to connect remote areas. Computer and network engineers must focus on deploying high-speed networks and configuring routing policies to ensure national data security and connectivity, enabling digital access across the country.
- **Digital Financial Inclusion:** The focus should be on developing secure, reliable digital services, such as **mobile banking applications**, which increase financial access and support inclusive economic growth in remote Nepalese villages.

- **E-Governance and E-Health Systems:** Implementing accessible digital platforms for governance (e-governance) and health services (e-health/telemedicine) would rapidly improve service delivery, reaching populations outside urban centers.
- **Cloud-Native Systems and Cybersecurity:** Engineers must design robust, secure, cloud-native systems for data storage and processing (e.g., photos stored in Google Drive) to support all digital infrastructure. Simultaneously, robust **cybersecurity systems** must be built to protect citizens' data and national digital systems, fostering trust in remote digital transactions.

Question 8: From the History of the engineering profession, what lessons would freshly graduated Engineers learn? What are the sectors that engineers in Nepal can practice their profession? (Marks: 5+5)

Part 1: Lessons from the History of Engineering (5 Marks)

By studying the history of engineering practice, freshly graduated engineers can learn several fundamental lessons that define their professional identity and responsibility:

1. **Engineering is Service to Mankind:** The primary lesson is that engineering is not just about technical tasks but is the application of scientific knowledge for the **benefit of mankind**. Historically, engineers have been the builders of civilization, providing the comforts and conveniences that society relies on.
2. **Specialization is a Product of Progress:** History shows that as societies evolved and were able to produce more food than they needed, individuals could spend time making "useful things," which led to the rise of specialized professions. Graduates must realize that their specialized technical skills are a contribution to this ongoing progress.
3. **Balance of Technical and Moral Values:** History highlights the difference between **Western values** (efficiency, progress, material comfort) and **Eastern values** (high morality, truth, and spiritual achievement). A fresh engineer learns that technical excellence must be balanced with moral orientation and ethical behavior.
4. **Awareness of Social Change:** Engineers have historically been agents of social change through mass production, automation, and communication. However, these changes bring both **merits** (higher living standards, women in technical fields) and **demerits** (mechanical life, weakening of family ties, urban slums). A graduate must learn to anticipate and mitigate these negative impacts.
5. **Adopting a Systematic Role:** History teaches the importance of the systematic engineering lifecycle: **Creating a Vision** (imagining products), **Preparing a Mission** (economic production), **Execution** (implementation), and **Monitoring** (ensuring accuracy and quality).

Part 2: Sectors for Engineering Practice in Nepal (5 Marks)

Engineers in Nepal have opportunities to practice their profession across various sectors of the economy:

Sector Type	Description and Examples
Public Sector (Government)	Organizations run by the government budget where any qualified citizen can compete through the Public Service Commission (PSC). This includes ministries like Physical Planning and Works, Water Resources, and Information and Communication.
Semi-Governmental Corporations	Autonomous bodies where the government holds significant influence. Examples include the **Nepal Electricity Authority (NEA)**, **Nepal Telecommunication Company (NTC)**, **Nepal Rastrya Bank**, and the National Construction Company of Nepal (NCCN).
Private Sector	An open market with thousands of organizations. This includes **Construction Companies** (A, B, C, and D classes), **Engineering Consulting Firms**, and manufacturing industries like Hetauda Cement.
IT and E-Business	A rapidly growing urban sector focused on **software design**, programming, and digital transformation. This also includes computer institutes and startups showcasing innovations at events like CAN InfoTech.
Academic and Research	Teaching and research opportunities in **Engineering Colleges** (both government like IOE and over 30 private colleges) as well as national bodies like the **Nepal Academy of Science and Technology (NAST)**.
Developmental Organizations (NGOs/INGOs)	Non-Governmental and International Organizations that require engineering expertise for rural infrastructure, health systems, and environmental projects.

Chapter - 2

Question: List the roles and responsibilities of the main professional engineering institutions in Nepal (NEC, NAST, NEA, NTA) that help you run your career smoothly.

In Nepal, several institutions govern and support the engineering profession, each playing a specific role in an engineer's career development:

Institution	Role and Responsibility
Nepal Engineering Council (NEC)	A statutory regulatory body that registers and licenses engineers to practice legally in Nepal. It sets educational standards, accredits engineering programs, enforces the professional Code of Conduct, and takes disciplinary actions (like suspending licenses) for unethical behavior.
Nepal Engineers' Association (NEA)	A voluntary professional organization representing engineers. It provides a platform for networking, organizes national conventions, conducts technical trainings and workshops, and promotes the rights and welfare of engineers.
Nepal Academy of Science and Technology (NAST)	Focuses on promoting research, innovation, and national scientific development. It supports engineers by funding or encouraging projects in fields like robotics, AI, and software innovation.
Nepal Telecommunications Authority (NTA)	While not strictly a professional body for individuals, it regulates ICT and telecommunications services. It impacts the careers of IT and computer engineers by setting national standards for digital infrastructure and services.

Question: As a member of Nepal Engineers' Association (NEA), what would you suggest to promote professional capacity of a fresh engineer?

To enhance the professional capacity of newly graduated engineers in Nepal, the following suggestions can be implemented:

- **Skill-Specific Workshops:** NEA should organize training on emerging technologies (e.g., AI frameworks, Cloud computing, or DevOps) to help fresh graduates stay relevant as technology evolves.
- **Mentorship Programs:** Establishing a formal network where senior professionals mentor juniors in both technical skills and ethical decision-making.
- **Industry Networking:** Creating more platforms, like student conferences or industry meetups, to provide fresh engineers with recognition and exposure to potential employers.
- **Continuing Education:** Offering certifications and access to global research journals to encourage lifelong learning.
- **Professional Orientation:** Providing workshops on labor laws, contract management, and building codes to prepare them for real-world site management and legal responsibilities.

Question: Discuss the ethical considerations related to the extensive use of artificial intelligence (AI) in the engineering field.

The rapid integration of AI in engineering introduces several ethical challenges that professionals must address:

- **Public Safety:** Engineers must ensure AI-driven systems (like autonomous traffic management) are thoroughly tested to prevent malfunctions that could endanger lives.
- **Algorithmic Bias:** There is a risk of developing biased algorithms, such as AI hiring tools that unfairly discriminate against certain demographics. Engineers have a duty to fix these biases to ensure fairness.
- **Privacy and Data Protection:** AI relies on massive datasets, often including sensitive personal information. Engineers must use encryption and strictly avoid unauthorized access to protect citizen data.
- **Liability and Accountability:** When an AI system fails, it is difficult to assign blame. Engineers must maintain high standards of care (diligence) in AI design to avoid "professional negligence".
- **Transparency:** Professionals must be honest about how AI models make decisions and must not hide bugs or vulnerabilities to meet deadlines.

Question: Define a profession and explain its characteristics/principles (or define professionalism and discuss its key characteristics).

Definition

A **profession** is a disciplined group of individuals who possess specialized knowledge and skills acquired through formal higher education and training, which they apply ethically to serve society. **Professionalism** is the combination of this specialized knowledge with high moral orientation and ethical behavior.

Key Characteristics of a Profession

Characteristic	Description
Specialized Knowledge	Grounded in a structured and complex knowledge system developed through research (e.g., data structures, architecture).
Formal Education	Requires prolonged academic preparation, typically a minimum of 3-4 years in a university.
Ethical Standards	Adherence to strict codes of conduct (like the NEC Code of Conduct) that guide responsible behavior.
Service Orientation	The primary duty is serving the public welfare and society responsibly rather than just earning.
Autonomy	Professionals are trusted to exercise independent judgment and make decisions based on their expertise.
Licensing/Regulation	Entry is restricted to qualified individuals registered with a formal professional body (e.g., NEC).
Lifelong Learning	A commitment to continuous improvement as knowledge in the field evolves rapidly.

Question: How do engineering and computer engineering fulfill the criteria of a profession? Illustrate with examples from Nepal.

Engineering fulfills professional criteria through its rigorous training and ethical mandates:

- **Advanced Education:** In Nepal, becoming an engineer requires a 4-year degree (8 semesters) with hands-on projects and laboratory work.
- **Legal Regulation:** Engineers in Nepal must register with the **Nepal Engineering Council (NEC)** to practice legally, ensuring only qualified people perform technical tasks.
- **Adherence to Ethics:** Engineers must follow the **NEC Code of Ethics**, which prioritizes public safety and honesty.
- **Example 1 (Computer Engineering):** Designing a secure e-voting system for Nepal's elections requires specialized expertise in encryption and user authentication that cannot be learned casually.
- **Example 2 (Civil/Structural):** Architects and engineers in Nepal must follow the **National Building Code** (e.g., seismic designs) to protect lives during earthquakes, reflecting a high responsibility toward public welfare.

Question: Explain moral dilemma and how it occurs. Explain ethical decision-making in engineering with examples. [1+3]

Moral Dilemma and How it Occurs

A **moral dilemma** is a situation where an individual must choose between two or more conflicting moral principles, each having a strong justification. It occurs when:

- Fulfilling one professional duty violates another (e.g., loyalty to an employer vs. truth to the public).
- There is uncertainty about the long-term consequences of a decision.
- There is professional or financial pressure to take shortcuts.

Ethical Decision-Making Process

Engineers use a structured 5-step process to navigate these dilemmas:

1. **Recognize the Issue:** Identify the conflict and the ethical principles involved.
2. **Gather Facts:** Clarify the situation without acting on assumptions.
3. **Evaluate Alternatives:** Consider legal obligations, codes of ethics, and the impact on stakeholders.
4. **Make a Decision:** Choose the path that best aligns with public welfare and integrity.
5. **Act and Reflect:** Implement the solution and later evaluate if it upheld professional standards.

Example: A software engineer in a Nepali ISP is asked to monitor employee emails without their consent. The dilemma is between following the employer's order and respecting privacy rights. The ethical choice is to refuse the request and inform the management that such surveillance violates user privacy.

Question: Define moral and immoral actions. Explain the perception of the general public towards Engineers and their professional works in the context of Nepal. [2+2]

Definitions

- **Moral Actions:** Standard behaviors that are accepted by the culture, religion, and social norms of a society. They are considered "right" behaviors.
- **Immoral Actions:** Behaviors that deviate from established moral standards and social values, often considered "wrong" or harmful.

Public Perception of Engineers in Nepal

- **High Status:** Generally, the public views engineering as a prestigious profession, and licensed engineers achieve high status and rewards for their specialized expertise.
- **Trust in Safety:** Society places significant trust in engineers to ensure that infrastructure (like bridges and buildings) is safe, especially given Nepal's earthquake risks.
- **Concerns over Professionalism:** There is sometimes a negative perception that professionalism is declining due to issues like low salaries, corruption in procurement, and a lack of political commitment to enforcing regulations.

Question: What are the secular systems or methods (fundamental laws of ethics) for making ethical proper and good decisions?

Secular ethical theories provide a framework for engineers to evaluate their choices based on reason rather than religious doctrine:

- **Utilitarian Law (Consequence-Based):** This principle states that one should choose the action that creates the **greatest benefit for the largest number of people**. It focuses purely on the outcome.
- **Universalism Law (Duty-Based / Deontology):** This law focuses on the **intent and moral duties** of the doer. It suggests that some actions are inherently right or wrong, regardless of their consequences (e.g., "always be honest").
- **Distributive Justice Law:** This is based on the **primacy of justice and equality for all**. Decisions should ensure that rules apply equally to everyone.
- **Personal Liberty Law:** This states that an act is unacceptable if it **violates anyone’s personal liberty**, even if that act would benefit a larger group of people.
- **Virtue-Based Approach:** This method asks, "**What would a good, honest, and responsible engineer do?**" focusing on the character and integrity of the professional.

Question 1: Define negligence, tort, and liability (or differentiate tort liabilities and vicarious liabilities). [1+3]

Definitions

- **Negligence:** It is the failure to exercise the level of care, skill, or diligence expected from a competent professional. It involves actions (or lack of actions) that are against established standards.
- **Tort:** A civil wrong that unfairly causes someone else to suffer loss or harm, resulting in legal liability for the person who commits the act. It is a wrongful act done willfully or negligently without a breach of contract.
- **Liability:** The legal or professional responsibility an engineer bears for errors, misconduct, or failure to fulfill their duties. It is the obligation to pay for damages or losses resulting from negligence or tort.

Differentiating Tort Liabilities and Vicarious Liabilities

Feature	Tort Liability	Vicarious Liability
Primary Actor	The individual who directly commits the wrongful act (the tortfeasor).	The subordinate or employee who commits the act during their job.
Responsibility	The individual is personally responsible for their own actions.	The employer (superior) is held responsible for the actions of the employee.
Context	Arises from a direct breach of a duty of care by a person.	Arises from the "right, ability, or duty to control" the activities of a violator.
Example	An engineer designs a faulty beam willfully.	A programmer at a firm writes poor code causing loss; the firm is held responsible.

Question 2: What are the conditions/elements necessary to prove professional negligence in engineering practice?

For negligence to be legally established and for a person to be compensated, the following four conditions must be satisfied:

1. **Duty of Care:** It must be proven that the engineer had a legal obligation to behave in a certain way towards the claimant.

2. **Breach of Duty:** The engineer failed to meet the expected professional standards or acted inappropriately.
3. **Causation:** The breach of duty must be the direct cause of the injury, loss, or damage.
4. **Damages:** The client or the public must have suffered an actual loss, injury, or damage as a result of the breach.

Question 3: What is meant by professional ethics? How do NEA and NEC contribute to maintaining the profession to remain ethical?

Meaning of Professional Ethics

Professional ethics refers to the moral principles and obligations that professionals must follow to maintain their professional status. It indicates how professionals ought to behave in their work, ensuring that practice is focused on social progress rather than just personal gain.

Contribution of NEA and NEC

- **Nepal Engineering Council (NEC):** As a statutory body, NEC maintains ethics by publishing a mandatory **Code of Conduct**, registering only qualified engineers, and taking disciplinary actions (suspending or revoking licenses) against those who violate ethical standards.
- **Nepal Engineers' Association (NEA):** As a voluntary professional organization, NEA contributes by providing a platform for sharing technical knowledge and innovations, conducting trainings, and focusing on safeguarding the rights and welfare of engineers so they can practice with dignity.

Question 4: Why is ethics/moral necessary in the engineering profession? Write and explain the bases of making ethical decisions. [4+6]

Necessity of Ethics/Morals

- **Public Safety:** Engineering decisions (like designing bridges or medical software) directly affect human lives.
- **Trust:** Society places great trust in engineers; ethical conduct builds and maintains this credibility.
- **Handling Sensitive Data:** Engineers often manage private data and national security systems which require high integrity.
- **Social Impact:** Engineers shape society and the environment; ethics ensures these impacts are positive and sustainable.

Bases of Making Ethical Decisions (Fundamental Laws)

1. **Eternal Law:** Based on nature and scriptures; it suggests there are obvious moral standards everyone should follow (e.g., "Do unto others...").
2. **Utilitarian Law:** Focuses on the outcome; the best decision is the one that creates the **greatest benefit for the largest number of people**.
3. **Universalism (Deontology):** Focuses on the **intent or motive**; it states that professionals have a duty to act with good motives regardless of the result.
4. **Distributive Justice Law:** Based on the principle of **equality**; rules and laws must apply equally to all people.
5. **Personal Liberty Law:** Any act that violates a person's **individual liberty** is unacceptable, even if it benefits a larger group.

Question 5: Explain the role of the Nepal Engineering Council (NEC) to control the immoral acts of engineers in Nepal.

The NEC controls immoral acts through several legal and administrative powers granted by the NEC Act:

- **Enforcement of Code of Conduct:** It mandates a set of rules (discipline, honesty, confidentiality) that all registered engineers must follow.
- **Licensing Power:** No engineer can legally practice without NEC registration, which allows the council to vet professionals.
- **Examining Committee:** A three-member committee can be formed to investigate any formal complaint registered against an engineer.
- **Disciplinary Action:** If found guilty of unethical or negligent behavior, the NEC has the authority to **suspend or revoke** an engineer’s license.
- **Accreditation:** By monitoring the quality of engineering colleges, NEC ensures that future engineers are trained under proper professional standards.

Question 6: "You should not practice Engineering without acquiring a license." Justify this statement.

This statement is justified for the following reasons:

- **Legal Requirement:** Under the Nepal Engineering Council Act, it is mandatory to be registered with the council to practice engineering legally in Nepal.
- **Public Safety:** Licensing ensures that only those with verified academic qualifications and skills perform technical tasks that could otherwise endanger lives.
- **Accountability:** A licensed engineer is personally responsible for their work and can be held liable for negligence.
- **Standardization:** It helps categorize engineers (General, Professional, Foreign) based on their expertise, ensuring the right person is doing the right job.
- **Enforceable Ethics:** Only registered engineers are bound by the professional Code of Conduct, which protects the public from malpractice.

Question 7: Recently an engineer was assaulted publicly by a group of people in Nepal. Explain the role that Professional Associations should play regarding the safety/welfare of engineers so that such incidents do not repeat.

Professional Associations (like NEA) should take the following roles in such scenarios:

- **Advocacy and Safeguarding Rights:** They must act as the "voice of the profession," demanding legal protection and justice for the assaulted member.
- **Legal Support:** Providing legal aid to ensure the perpetrators are punished according to the law of the land.
- **Liaison with Government:** Working with ministries and law enforcement to create a safer working environment for technical personnel.
- **Promoting Professional Dignity:** Organizing public awareness campaigns to highlight the critical role of engineers in national development and safety.
- **Policy Formulation:** Suggesting amendments to labor and security laws to provide specific protection for site-based professionals.

Question 8: What are the codes of ethics for engineers according to the Nepal Engineering Council (NEC)? Describe the code of ethics for engineers generally.

NEC Professional Code of Conduct

1. **Discipline and Honesty:** Service must be conducted in a disciplined and honest manner.
2. **Politeness and Confidentiality:** Information must remain confidential unless client consent or law requires otherwise.
3. **Non-discrimination:** No discrimination based on religion, race, sex, or caste.
4. **Field of Competence:** Engineers should only work within their area of study and expertise.
5. **No Improper Financial Gain:** No obtaining bribes or improper gains beyond salary and benefits.
6. **Personal Responsibility:** Engineers are personally responsible for all their professional work.
7. **Clear Identification:** Must state name, designation, and registration number on all professional documents.

General Engineering Ethics

- **Public Welfare:** The safety, health, and welfare of the public are paramount.
- **Honesty:** Issuing public statements only in a truthful and objective manner.
- **Faithful Agent:** Acting as a faithful agent for employers/clients and avoiding conflicts of interest.

- **Reputation:** Building a reputation on merit and not competing unfairly with others.

Question 9: How should engineers respond to professional dilemmas in ICT and computer engineering fields?

Engineers should use a structured **Ethical Decision-Making** approach:

1. **Recognize the Issue:** Identify if a situation involves conflicting values (e.g., privacy vs. profit).
2. **Gather Facts:** Clarify all technical and legal details before acting; avoid assumptions.
3. **Evaluate Alternatives:** Look at the **NEC/IEEE/ACM Code of Ethics**. For example, if tasked to install hidden tracking software, the engineer should recognize this violates user privacy.
4. **Prioritize Stakeholders:** Use the priority list: **Society > Profession > Organization > Individual**.
5. **Make an Ethical Choice:** Choose the path that protects the public. In ICT, this often means fixing security vulnerabilities immediately, even if it delays a product launch.
6. **Take Action:** Refuse unethical requests (like stealing data or software piracy) and report them if public safety is at risk.

Question 10: Differentiate between NEA and NEC.

Feature	Nepal Engineering Council (NEC)	Nepal Engineers' Association (NEA)
Nature	Statutory/Autonomous Regulatory Body (Established by Act).	Voluntary Professional Organization/NGO.
Primary Goal	To regulate the profession and protect the public.	To safeguard the rights and welfare of engineers.
Function	Licensing, accreditation of colleges, and disciplinary actions.	Networking, training, workshops, and sharing research.
Membership	Registration is **mandatory** to practice legally.	Membership is **voluntary**.
Code Focus	Enforces the Professional Code of Conduct legally.	Promotes ethical standards through professional development.

Chapter - 7

Case Study 1: Ethical Dilemma (Hazardous Waste Leakage)

Question: A factory has a leakage of hazardous chemical waste that is harmful to health and the environment. The boss requests the engineer to stay silent about the issue or face losing their job. Discuss what you would do.

Analysis of the Dilemma

This is a classic **moral dilemma**, where an individual must choose between two conflicting values: personal livelihood (job security) and public safety/environmental protection.

Priority of Obligations

In engineering, there is a clear hierarchy of responsibility that guides decision-making:

1. **Society (Public Safety/Environment)** - Highest Priority
2. **Profession**
3. **Organization (Employer)**
4. **Individual (Self-interest)** - Lowest Priority.

Steps I Would Take

1. **Recognize the Issue:** The leakage violates the **Nepal Engineering Council (NEC) Code of Ethics**, which mandates that engineers prioritize public safety and environmental responsibility.
2. **Gather Facts:** I would document the extent of the leakage, the specific hazards involved, and the potential impact on the surrounding community.
3. **Negotiate with the Boss:** I would first try to convince the boss to fix the issue. I would explain the legal risks (Criminal and Civil Liability) the company faces if the leakage is discovered later or causes an accident.
4. **Whistleblowing:** If the boss refuses to act and continues to pressure me to stay silent, I have a moral and professional duty to report the event to the concerned authorities (such as the Ministry of Environment or local government).
5. **Conclusion:** Protecting the health of thousands of people and the environment is far more important than a single job. Staying silent would constitute **Professional Negligence** and could lead to the revocation of my engineering license.

Case Study 2: Liability and Negligence (Structural Damage)

Question: A number of cracks, structural as well as settlement, appeared in a building designed by an engineer within two years of its completion. It was found that no soil investigation was done, and the reinforcement report was prepared during construction. Who is ultimately responsible (Owner, Engineer, or Municipality)? Discuss the roles.

Responsibility Matrix

Party	Role and Responsibility	Liability Assessment
Engineer	Must ensure technical precision, perform soil analysis, and supervise construction according to codes.	**Primary Responsibility.** Designing without a soil report is **Gross Professional Negligence**. Signing reports after construction starts is a breach of the "Duty of Care".
Owner	Responsible for funding and ensuring they hire competent professionals who follow regulations.	**Secondary/Contributory Liability.** If the owner pressured the engineer to skip the soil test to save money, they share blame for **Contributory Negligence**.
Municipality	Responsible for enforcing building bylaws and the National Building Code (NBC).	**Regulatory Failure.** The municipality is responsible for failing to audit the design documents properly before issuing a building permit.

Judgment

The **Engineer** is ultimately the most responsible for the structural failure. Under the **NEC Act**, a professional is personally responsible for their professional work. Designing a building without a soil test is a direct violation of standard engineering practice and constitutes a breach of duty that caused actual damages.

Case Study 3: Ethics and Conflict of Interest (Friend/Accommodation)

Question: You are a consulting engineer and your friend (a contractor on the same project) invites you for dinner and offers his guest house for accommodation. What would you do? Prepare a case study.

Case Study Facts

- **Parties:** Consulting Engineer (The Evaluator) and Contractor (The Implementer).
- **Relationship:** Personal friendship.
- **Event:** Offer of free accommodation in a private guest house.

Ethical Analysis

As per the **ABET** and **NEC Code of Ethics**, an engineer must act as a **"faithful agent"** for their employer or client and strictly avoid **Conflicts of Interest**.

- **Conflict of Interest:** Accepting favors (free accommodation) from a contractor while supervising their work could influence—or appear to influence—the engineer's professional judgment regarding quality control.
- **Undue Influence:** The guest house offer could be seen as an attempt to create a "debt" that the contractor might later use to get substandard work approved.

My Decision

1. **Refusal of Gift:** I would politely decline the offer of the guest house. I would explain that my professional code of conduct prohibits me from accepting gifts or personal favors from contractors on the same project.
2. **Maintaining Boundaries:** While I might accept a simple social dinner as a friend, I would ensure that no project-related talk occurs and that all official project interactions remain transparent.
3. **Transparency:** I would disclose the friendship to my employer (the client) to maintain "Clean Hands" and ensure there are no surprises.

Case Study 4: Tort, Liability, and Negligence (Sub-contractor Beating)

Question: A sub-contractor was severely beaten by his laborers because he had not paid them for seven months. Investigation revealed the client did not pay the contractor/sub-contractor because the consultant failed to submit the necessary bill reports. Discuss the case considering contract law, tort, liability, and negligence.

Legal Breakdown

Legal Concept	Application to the Case
Contract Law	The consultant has a **Breach of Contract** with the client for failing to perform their duty (submitting bill reports), which blocked the payment chain.
Tort	The beating of the sub-contractor is a **Tort** (specifically assault/battery). While the consultant didn't do the beating, their actions directly led to the environment of suffering.
Liability	The Consultant bears **Professional Liability** for the delay. The Client might face **Vicarious Liability** for the consultant's failure if they are the consultant's employer.
Negligence	The consultant's failure to submit reports for 7 months is **Gross Negligence**, as it shows a severe lack of diligence and disregard for the welfare of workers.

Discussion

The **Consultant** is the "root cause" of the conflict. By failing their professional duty, they caused financial harm down the entire chain. The **Laborers** committed a criminal act (physical assault), but the **Sub-contractor** is a victim of both the beating and the consultant’s negligence. The Consultant should be held liable for the financial losses and face disciplinary action from the NEC.

Case Study 5: Ethical Dilemma (Unwanted Bargain/Political Pressure)

Question: An engineer acting as a political leader pressures a contractor for low-quality construction and bargains for unwanted payment. Write for and against this.

Arguments AGAINST (The Professional & Ethical View)

- **Violation of NEC Code:** The NEC Code of Conduct explicitly forbids obtaining **improper financial gain** or performing acts that impair the dignity of the profession.
- **Public Safety:** Pressuring for "low-quality construction" is a direct threat to human life and violates the foremost duty of an engineer: protecting the public.
- **Corruption:** This behavior is a form of corruption that harms national development and creates **Professional Negligence**.

Arguments FOR (The "Hypothetical" Political Reality)

- *Note: This is purely for analytical purposes; there is no ethical justification.*
- **Resource Mobilization:** A political leader might argue they need funds for "party mobilization" or social work.
- **Project Feasibility:** They might argue that reducing quality is the only way to finish a project within a very low government budget (this is technically flawed and dangerous) [Self-knowledge].

Final Conclusion

There is **no professional justification** for this behavior. An engineer-politician is still an engineer and must uphold the **Dignity of the Profession**. They can be legally prosecuted for fraud and have their license revoked by the NEC.

Case Study 6: Negligence Judgment (RCC Bridge Failure)

Question: An RCC bridge develops cracks after 6 months. Investigation shows: 1. Design was okay. 2. Steel used was not tested. 3. Workmanship was not as per specification. 4. Supervision by the consultant was lacking. 5. Traffic exceeded the 20-ton design load. Provide your judgment.

Judgment of Failure

The failure is a result of **Multiple Failures** across all parties:

1. **Consultant's Negligence:** They failed their duty of **Supervision** and allowed workmanship to decline and untested steel to be used. This is **Gross Professional Negligence**.
2. **Contractor's Breach:** They are responsible for **Breach of Contract** for using untested materials and failing to meet workmanship specifications.
3. **Owner/Authority's Liability:** They are responsible for the **Causation** factor of overloaded traffic. If the bridge was designed for 20 tons and the owner allowed heavier vehicles, they share liability for the damage.

Conclusion

While the overload was a trigger, the bridge was already compromised by **poor materials and workmanship**. The Consultant and Contractor are **jointly liable** for the technical failure, while the Owner is responsible for the operational failure (overloading).

Chapter - 3

Question 1: Elaborate on the detailed duties/terms of reference (TOR) or job descriptions of fresh graduate Engineers (specifically Computer Engineers) working in the public and private sectors in Nepal.

In Nepal, the job descriptions for fresh engineering graduates differ slightly based on whether they are employed in a government-funded public office or a business-oriented private firm.

Job Description in the Public Sector

Engineers in the public sector are typically recruited through the Public Service Commission (PSC) and work within various ministries or government projects. Their duties focus on national infrastructure and service delivery:

- **Survey and Design:** Carrying out preliminary and detailed technical surveys, designs, and cost estimates for projects.
- **Project Execution:** Directly executing project works or assigning specific tasks to subordinates for implementation.
- **Capacity Building:** Conducting programs aimed at increasing the skills and technical capacity of the people or local community.
- **Technical Reporting:** Preparing various formal documents, including feasibility reports, progress reports, monitoring reports, and final completion reports.
- **Supervision:** Monitoring and evaluating ongoing project activities to ensure they meet government standards.
- **Agency Coordination:** Facilitating and coordinating with donor agencies involved in project funding or technical support.
- **Administrative Tasks:** Performing other technical or administrative jobs as assigned by immediate superiors.

Job Description in the Private Sector

In private organizations, such as construction firms or software houses, the focus is more on coordination, quality control, and efficient management.

- **Stakeholder Coordination:** Acting as a bridge to coordinate work between clients, consultants, and contractors.
- **Site Management:** Laying out works on-site, surveying, and providing accurate estimations for resources.
- **Work Supervision:** Supervising, monitoring, and controlling the daily progress of works to meet deadlines.
- **Quality Control:** Assessing the quality of workmanship and materials, and reporting these assessments to the relevant authorities.
- **Financial Documentation:** Preparing bills as a quality surveyor and maintaining technical records.
- **Project Planning:** Planning the project timeline and providing regular progress reports to management.

- **Claims and Reports:** Preparing detailed technical reports and handling any technical claims that arise during the project.
- **Staff Training:** Providing necessary technical training regarding site operations and office systems to new staff members.

Comparison of Duties

Feature	Public Sector Focus	Private Sector Focus
Core Goal	Public welfare and policy implementation.	Economic investment and efficient production.
Recruitment	Public Service Commission (PSC).	Direct hiring by entrepreneurs/firms.
Primary Task	Feasibility, survey, and community capacity.	Coordination, quality control, and bill preparation.
Authority	Derived from government mandates.	Derived from project-specific stakeholders.

Question 2: What are the sectors in Nepal where engineers can practice their profession?

Engineers in Nepal have a wide range of opportunities across different sectors of the economy:

1. **Public Sector:** This includes all government ministries, departments, and regional or district offices funded by the national budget. It involves large-scale infrastructure, policy planning, and public service.
2. **Semi-Governmental Corporations:** These are autonomous bodies or companies where the government holds majority shares. They often handle strategic services like electricity, telecommunications, and banking.
3. **Private Sector:** This is an open market consisting of thousands of private organizations, including construction companies, consulting firms, and industrial manufacturing units.
4. **Information Technology (IT) and E-Business:** A rapidly growing sector in urban centers like Kathmandu, involving software design, programming, and digital transformation projects.
5. **Academic Sector:** Both government and private engineering colleges affiliated with various universities employ engineers for teaching and research.
6. **Non-Governmental Organizations (NGOs/INGOs):** Many developmental projects run by national and international NGOs require technical engineering expertise.

Question 3: List at least 5 organizations each in the public and private sectors in Nepal where engineers practice.

Public/Semi-Governmental Organizations

1. **Nepal Electricity Authority (NEA):** Responsible for power generation, transmission, and distribution.
2. **Nepal Telecommunication Company (NTC):** The primary provider of telecommunication services across the country.
3. **Ministry of Physical Planning and Works:** Handles national infrastructure development, including roads and bridges.
4. **Nepal Rastrya Bank:** Employs engineers for technical infrastructure and security systems.
5. **Tribhuvan University (Institute of Engineering):** Employs engineers in academic, research, and technical roles.

Private Sector Organizations

1. **United Builders and Company:** A prominent A-class construction company in Nepal.
2. **Private Engineering Colleges:** Numerous private institutions affiliated with universities that employ technical faculty.
3. **Software Development Firms:** Companies in Kathmandu and other cities focusing on software design and IT services.
4. **Engineering Consulting Firms:** Many private firms that provide specialized design and feasibility services.
5. **Manufacturing Industries:** For example, private cement companies or hydropower developers operating in the open market.

Chapter - 4

Question 1: Discuss the importance, essentials (elements), and types of a valid contract.

Meaning and Definition of a Contract

A contract is a formal agreement between two or more competent parties where an offer is made and accepted, providing benefits to each involved party. According to the Contract Act 2056 of Nepal, a contract is an agreement enforceable by law for performing or not performing any work.

Importance of a Contract

Contracts are vital in professional engineering for several reasons:

- **Legal Enforceability:** It makes an agreement legally binding, allowing parties to seek legal remedies if the agreement is breached.
- **Defining Duties:** It clearly specifies what the contractor or service provider must do and what the owner must pay.
- **Quality and Quantity:** It sets the standards for the quantity and quality of work to be delivered.
- **Timeframe:** it specifies the exact duration for project completion and the schedule for payments.
- **Risk Management:** It identifies the representatives of each party and sets out courses of action for different potential situations and risks.
- **Termination Procedures:** It outlines the process for how a contract can be legally ended.

Essentials (Elements) of a Valid Contract

For a contract to be legally valid and enforceable, it must contain these essential elements:

1. **Offer and Acceptance:** One party must make a promise (offeror), and the other must give consent (offeree) in the same spirit.
2. **Two or More Competent Parties:** The parties involved must be legally recognized entities.
3. **Capacity to Contract:** Parties must be of legal age (above 16 in Nepal) and of sound mind (not insane or under the influence).
4. **Mutual Intent (Legal Relationship):** There must be a clear intention by both parties to establish a formal legal relationship rather than just a social agreement.
5. **Consideration:** This is something of value exchanged between the parties (e.g., money for services).
6. **Lawful Purpose:** The objective of the contract must not be illegal or against the law of the land.
7. **Free Consent:** The agreement must be made without coercion, undue influence, fraud, or misrepresentation.

8. **Possibility of Performance:** The task agreed upon must be physically and legally possible to perform.
9. **Certainty:** The terms and conditions must be clear and not vague.
10. **Written and Registered:** While some contracts can be verbal, professional engineering contracts should be written and often registered for legal validity.

A **Valid Contract** is one that complies with all the essentials mentioned above and is binding and enforceable on all parties involved.

Question 2: List down various methods of work execution and explain any three mostly practiced types of contract briefly.

Methods of Work Execution

In the context of public procurement in Nepal, various methods are used for executing works or procuring goods:

1. International Contract Bidding (ICB)
2. National Contract Bidding (NCB)
3. Sealed Quotations
4. Direct Procurement (Direct Purchase)
5. Users' Committee or Beneficiary Group
6. Force Account (Amanat)

Three Mostly Practiced Types of Contract

Type of Contract	Brief Explanation
Unit Price/Rate Contract	Used when the exact quantity of work is difficult to determine beforehand. Payment is made based on the actual measurement of work units completed in the field at an agreed-upon rate per unit.
Lump Sum Contract	Often used for clearly defined assignments like feasibility studies or project designs. The consultant or contractor agrees to perform the entire specified assignment for a single, fixed total price.
Sealed Quotations	A simplified procurement method used for lower-value works (up to 2 million NPR for construction). It requires inviting quotes from at least three firms with a 15-day notice period.

Question 3: Differentiate between void and voidable contracts (or in which condition a contract will be void).

Feature	Void Contract	Voidable Contract

Definition	A contract that is invalid from the moment it is created (void ab initio).	A valid contract that can be declared invalid by one of the parties for specific legal reasons.
Enforceability	It is nonexistent in the eyes of the law and cannot be upheld.	It is binding to at least one party but can be repudiated (cancelled) by the aggrieved party.
Performance	No performance of work is legally possible or required.	Performance is possible unless and until the contract is cancelled.
Reasons	Illegal activities, impossible tasks, or improper structure.	Coercion, fraud, undue influence, or contracts involving minors.

Conditions under which a contract will be Void:

A contract is considered void if it:

- Prevents someone from engaging in a lawful profession or trade.
- Restrains marriage (except where prohibited by law).
- Seeks to prevent the enforcement of legal rights in court.
- Is contrary to public morality or public interest.
- Involves an impossible job or task.
- Is signed by an incompetent person (e.g., minor or insane person).
- Involves an unlawful objective or consideration.
- Has a vague meaning that does not provide reasonable understanding.

Question 4: As an Engineer, how does knowledge of contract help you in your professional career?

Knowledge of contracts is essential for an engineer's professional practice because:

- **Legal & Ethical Protection:** It helps engineers avoid legal trouble and understand their liabilities for errors or misconduct.
- **Informed Decision Making:** Understanding contractual obligations allows engineers to make responsible decisions that protect public safety and the client's interests.
- **Defining Relationships:** It clarifies the professional relationship and boundaries between the engineer, client, and contractor.
- **Ensuring Quality:** By understanding specifications and standards set in contracts, engineers can better supervise work to prevent system failures or data breaches.
- **Dispute Resolution:** Knowledge of contract terms helps in settling disagreements through established methods like adjudication or arbitration.
- **Career Credibility:** Acting according to contractual and ethical standards builds trust with society and enhances an engineer's reputation and prospects.

Question 5: Differentiate between the bid validity period and bid security validity period.

Feature	Bid Validity Period	Bid Security Validity Period
Purpose	The duration for which the bidder's offer/price remains binding and cannot be changed.	The duration for which the earnest money/guarantee provided by the bank remains active.
Duration (NCB)	Typically **90 days** (for projects up to 10 crore) or **120 days** (above 10 crore).	Typically **120 days** or **150 days** (usually 30 days longer than the bid validity).
Duration (Sealed Quotation)	**45 days**.	**75 days**.
Purpose	Prevent price change	Protect employer

Question 6: Differentiate between contract and agreement.

Feature	Agreement	Contract
Nature	It is the simple acceptance of an offer (proposal).	It is an agreement that is specifically enforceable by law.
Legal Obligation	Does not necessarily create legal obligations between parties.	Creates a "perfect obligation" and legal bond between parties.
Scope	Every contract is an agreement, but not every agreement is a contract [My own knowledge/context of 216-217].	It is a narrower, legally defined subset of agreements.

Question 7: Define tender and describe the purpose of tendering. Why is it so important?

Definition of Tender

A tender (or bid) is a written offer by a contractor or supplier to execute a specific work or supply goods at a certain rate and within a fixed timeframe under defined conditions of agreement. It is considered the first step in the formulation of a contract.

Purpose and Importance of Tendering

- **Selecting the Best Party:** It helps the owner find the most competent and capable contractor or firm for a project.
- **Economic Efficiency:** It ensures "best value for money" by allowing for competitive pricing among eligible bidders.
- **Transparency:** It maintains a fair and transparent procurement process at all stages.
- **Equal Opportunity:** It gives all eligible bidders a fair and equal chance to compete for a project.
- **Quality Control:** It helps in achieving quality outputs by evaluating technical proposals against set standards.

Question 8: List the essential information to be furnished/enclosed in a tender notice when publishing for a public purpose.

A tender notice published for public purposes should include:

1. **Authority Name:** The name and address of the public entity inviting the bid.
2. **Publication Date:** The first date the notice is published.
3. **Project Description:** A brief description of the job, nature of work, and its location.
4. **Duration:** The estimated time for the project to be completed.
5. **Document Availability:** Where, when, and how the bid documents can be obtained.
6. **Cost:** The price of the tender/bid document.
7. **Submission Details:** The deadline, place, and manner for submitting the bids.
8. **Bid Security (Earnest Money):** The required amount (usually 2-3% in Nepal) and its validity period.
9. **Performance Security:** Information regarding the security deposit required from the successful bidder.
10. **Opening Details:** The date, time, and location where bids will be opened.
11. **E-Bidding Provision:** Details on whether electronic submission is allowed and the process for it.
12. **Acceptance Date:** The expected date when the successful bid will be accepted.

Question: Explain briefly procurement types (Sealed bidding, sealed quotation, and direct purchase/procurement) of work as per the Public Procurement Act 2063.

In Nepal, the **Public Procurement Act 2063** regulates how public entities acquire goods and construction works. The following table summarizes the three common types of procurement methods used for construction works:

Feature	Direct Purchase (Procurement)	Sealed Quotation	Sealed Bidding (NCB/ICB)
Monetary Limit (Works)	Up to NPR 500,000.	Up to NPR 2,000,000.	Above NPR 2,000,000.

Notice Period	No formal public notice required.	Minimum 15 days in national or local newspaper.	30 days (National) or 45 days (International).
Selection Criteria	Directly from one supplier or minimum 3 quotations if above 25,000.	Minimum of 3 quotations required; lowest cost is approved.	Competitive bidding; selection based on "best value for money".
Bid Validity	N/A	45 Days.	90 or 120 Days.

- **Direct Purchase/Procurement:** This is the simplest method where services are procured directly from a supplier without a complex bidding process, typically used for small repairs or urgent needs.
- **Sealed Quotation:** This involves inviting price statements in sealed envelopes from interested firms based on a prepared form with clear specifications. Once submitted, these cannot be amended or withdrawn.
- **Sealed Bidding (National/International):** Known as National Competitive Bidding (NCB) or International Competitive Bidding (ICB), this method is used for larger projects to ensure transparency and equal opportunity for all eligible bidders.

Question: What are the methods of recruitment of consultants? Explain the purpose of Expression of Interest (EOI) and Request for Proposal (RFP). [2+2]

Methods of Recruitment of Consultants

According to the Public Procurement Act, a public entity can procure consultancy services using two primary methods:

1. **Requesting Competitive Proposals:** Inviting different firms to submit their technical and financial offers.
2. **Direct Negotiations:** Engaging directly with a specific consultant in special circumstances.

Purpose of EOI and RFP

- **Expression of Interest (EOI):** The purpose of an EOI is to identify and shortlist capable and interested firms before inviting detailed bids. It requires a 15-day public notice and focuses on the firm's experience and capacity.
- **Request for Proposal (RFP):** The purpose of an RFP is to invite the shortlisted firms to provide a detailed technical and financial offer. It typically requires a 30-day notice period and serves as the basis for the final selection of the consultant.

Question: What are the preparations to be made before inviting a competitive bidding notice?

Before a public entity can publicly publish a tender or competitive bidding notice, several administrative and technical preparations are mandatory:

1. **Project Preparation:** The overall scope and objectives of the project must be defined.
2. **Specifications and Designs:** Detailed technical specifications, plans, drawings, and designs pertaining to the work must be finalized.
3. **Estimation of Quantities:** A Bill of Quantities (BOQ) must be prepared to list all items of work required.
4. **Cost Estimation:** A formal cost estimate must be prepared based on current rates.
5. **Administrative Approval:** The cost estimate and the project itself must receive approval from the concerned authorities.
6. **Resource Planning:** The entity must ensure that adequate funding and resources are available for project implementation.
7. **Bid Document Preparation:** Standard Bidding Documents (SBDs) must be prepared and approved, including instructions for bidders and conditions of the contract.

Question: How can a contract be interpreted in case of ambiguity? Explain the rules of contract interpretation.

Contract interpretation is the process of determining the exact meaning of the terms agreed upon by the parties. When ambiguities arise, the following rules and principles are applied:

- **Letter vs. Spirit of the Law:** Professionals are encouraged to "read between the lines" to understand the true intent of the documents as they were formulated by the parties.
- **Order of Priority:** If there is a contradiction between different documents in a contract, the established "Priority of Contract Documents" is used to decide which document prevails.
- **Requirement of Certainty:** According to the Contract Act 2056, a contract with vague or unclear terms that do not provide a reasonable meaning can be considered void.
- **Foresight:** In preparing contracts, parties are advised to use foresight to define words clearly and establish common meanings to prevent future disputes.
- **Consistency:** Contract language must remain consistent throughout all sections to avoid conflicting interpretations.

Question: What is a contract document and what are its priorities?

Meaning of a Contract Document

A contract document is a collection of papers that describe in detail the scope of the agreement, the technical specifications of the work, and the legal responsibilities of the parties involved. It serves to guide the procedure, pricing, and payment for the work during implementation.

Priority of Contract Documents

In the event of a dispute or ambiguity between different parts of a contract, the documents are typically ranked in the following order of priority:

1. **The Contract Agreement:** The formal signed agreement.
2. **The Letter of Acceptance:** The document notifying the bidder that their offer was accepted.
3. **The Bid / Tender:** The original offer submitted by the contractor.
4. **Conditions of Contract:** Both General and Special conditions.
5. **Detail Specifications:** The technical requirements for materials and workmanship.
6. **Detail Drawings:** The visual plans for the construction.
7. **Priced Bill of Quantities (BOQ):** The list of work items and their agreed rates.
8. **Addenda:** Any additional documents or clarifications issued to correct or provide more information.

Chapter - 5

Question 1: List characteristics, advantages, and disadvantages of business enterprises: Sole proprietorship, Partnership firm, and Limited company. [3+3]

The following table summarizes the key features of the three main types of business organizations practiced in Nepal:

Feature	Sole Proprietorship (Sole Business)	Partnership Firm	Limited Company (Private/Public)
Characteristics	Owned and managed by a single individual. Unlimited liability for the owner. Voluntary origin and end.	Joint ownership by two or more people (shared profit/loss). Established via agreement. Unlimited liability for partners.	Legal artificial person with perpetual existence. Limited liability for shareholders. Capital collected through shares.
Advantages	Easy to form and dissolve. Complete control and prompt decision-making by the owner. All profits belong to the owner.	More capital compared to sole business [Self-knowledge]. Shared risk among partners. Collective effort and shared skills.	Limited liability protects personal assets. Can raise huge capital by issuing shares. Perpetual existence (business continues if a member dies).
Disadvantages	Limited capital and resources. Unlimited liability (personal assets at risk). Lack of specialization and high risk of loss if owner is absent.	Possibility of misunderstanding or friction between partners. Unlimited liability. Difficulty in transferring ownership.	Difficulty and high cost in formation (many legal formalities). Lack of secrecy as financial statements are public/shared. Conflict of interest between shareholders and management.

Question 2: Discuss the Intellectual Property Rights (IPR) and its related policies and regulations. Why is it necessary?

Meaning and Policies

Intellectual Property Right (IPR) is the legal right one has over the creations of their mind, such as inventions, literary/artistic works, symbols, names, and designs. In Nepal, these rights are governed by the **Copyright Act 2059** and the **Patent, Design, and Trademark**

Act 2022. These laws provide the framework for the protection of artistic work (copyright) and industrial/commercial properties (patents and trademarks).

Why IPR is Necessary

- **Ownership and Protection:** It allows creators to own and protect their ideas just like physical property.
- **Legal Recourse:** It provides the right to initiate legal action against anyone using the work without permission.
- **Incentive for Innovation:** By granting exclusive rights, it encourages individuals to invest time and resources into new inventions and creative works.
- **Commercial Value:** It allows creators to control the sale, distribution, and adaptation of their work for a specified period, ensuring they benefit financially.

Question 3: Define trademark, service mark, patent, industrial design, and copyright in detail.

- **Trademark:** A symbol, word, phrase, logo, or design used to identify and distinguish the goods of one firm from another. It acts as a source indicator for customers.
- **Service Mark:** Similar to a trademark, but used to identify and distinguish services rather than physical products (e.g., logos of airlines or banks).
- **Patent:** A right granted for a new and useful invention, machine, or process. It gives the inventor exclusive control to prevent others from making, using, or selling the invention without permission for a period (usually 20 years).
- **Industrial Design:** Protects the visual, aesthetic, or ornamental features of an object (shape, pattern, or color) rather than its functional aspects. It can be 2D (lines/colors) or 3D (shape/surface).
- **Copyright:** The exclusive right granted to the creator of literary or artistic works (books, music, software, paintings) to produce, reproduce, or publish their work. It generally lasts for the life of the author plus 50 years.

Question 4: Explain the process of registering copyrights and a company in Nepal. What is the penalty for violating copyright?

Process of Registering a Company in Nepal

An application must be submitted to the **Registrar of Company** with the applicable fee and the following documents:

1. **Memorandum of Association:** Outlining the company's objectives and structure.
2. **Articles of Association:** Internal rules for managing the company.
3. **Joint Venture Agreement:** If applicable.
4. **Citizenship Certificates:** Copies of the certificates belonging to the founder stakeholders.

Process of Registering Copyrights

- In Nepal, protection starts as soon as the work is created.
- The creator can register their work at the **Nepal Copyright Registrar's Office** by providing proof of original authorship.

Penalty for Violating Copyright in Nepal

As per the Copyright Act 2059, the penalties for infringement are:

- **First offense:** A fine of NPR 10,000 to 100,000, or 6 months imprisonment, or both.
- **Subsequent offenses:** A fine of NPR 20,000 to 200,000, or 1 year imprisonment, or both.

Question 6: Explain various provisions regarding working hours, leave, and holidays, as well as health and safety provisions according to the Labor Act of Nepal.

Working Hours, Leave, and Holidays

- **Standard Hours:** 8 hours per day or 48 hours per week.
- **Rest Periods:** Workers cannot be forced to work more than 5 hours continuously; a 30-minute break for "Tiffin" and rest is mandatory.
- **Overtime:** Paid at 1.5 times the normal wage. Overtime cannot exceed 4 hours a day or 20 hours a week and cannot be forced.
- **Leave:** Provision for public holidays, sick leave, home leave, maternity leave, and mourning leave.

Health and Safety Provisions

- **Workplace Environment:** Must be clean, with adequate ventilation, light, and a suitable temperature.
- **Sanitation:** Separate modern toilets for males and females and hygienic drinking water must be provided.
- **Safety Equipment:** Necessary personal protective equipment (PPE) must be provided to avoid adverse health effects.
- **Pollution Control:** Proper waste disposal and protection from dust, noise, and fumes.
- **Medical Checkups:** Compulsory for workers at least once a year.
- **Fire Safety:** Fire exits and extinguishers must be present.

Question 7: Explain why engineers should be aware of labor law while working in public or private organizations.

Engineers must understand labor laws because:

- **Management Responsibility:** Engineers often act as project managers or site supervisors, making them responsible for the welfare and safety of workers.

- **Worker Efficiency:** Understanding and implementing welfare activities influences workers to work with more interest and efficiency.
- **Legal Compliance:** Engineers must ensure their site practices (like working hours and safety) comply with the **Labor Act 2048** to avoid legal penalties and stop-work orders.
- **Safety Accountability:** Knowledge of health and safety standards helps engineers prevent accidents, which can lead to severe legal and professional liability.

Question 8: What will you do if a construction worker dies by falling from a height on your site?

Note: While a specific step-by-step checklist for this scenario is not explicitly listed in the sources, the following response is derived from source information regarding engineer duties, labor act welfare, and legal liability.

In the event of a worker's death, an engineer should follow these steps:

1. **Immediate Action:** Stop all work at the site immediately to prevent further accidents and preserve the scene for investigation.
2. **Notification:** Inform the local police and the concerned labor office as required by law. Inform the employer and the family of the deceased [Self-knowledge].
3. **Emergency Care:** Ensure any other injured persons receive immediate medical attention.
4. **Internal Investigation:** Review if safety precautions (like safety harnesses or railings) were in place and if the "Duty of Care" was breached.
5. **Documentation:** Gather facts and document the conditions of the site at the time of the accident to avoid "Negligence" claims based on assumptions.
6. **Compensation:** Facilitate the process for the family to receive compensation for injury or loss of life as mandated by the Labor Act.

Question 9: How can building codes help people to be safe?

Building codes ensure safety through several regulatory mechanisms:

- **Minimum Standards:** They specify the minimum acceptable level of safety for constructed objects, ensuring they can withstand normal loads.
- **Disaster Resistance:** The **National Building Code (NBC)** in Nepal includes specific guidelines for seismic design to help buildings withstand earthquakes.
- **Fire and Life Safety:** Provisions in the code address fire safety, exits, and electrical requirements to prevent accidents.
- **Structural Integrity:** For "Professionally Engineered Buildings," the code mandates structural analysis and supervision by qualified engineers to prevent collapses.

- **Rule of Thumb:** In remote areas, "Mandatory Rules of Thumb" provide simple guidelines for small buildings to ensure they are safe even when professional engineers aren't available.
- **Planned Urbanization:** Building bylaws ensure adequate "Right of Way" and setbacks to allow for emergency services (like fire trucks) to access buildings.

Chapter - 6

Question 1: Conflict is said to be an inevitable outcome of behavioral interactions. In your opinion, is it sensible to have conflict or should it be avoided as far as possible? Justify your answer.

Conflict is a natural and inevitable part of any organization or engineering project because it involves groups of people with different goals, roles, and perceptions.

Traditional vs. Modern View of Conflict

- **Traditional Approach:** Older views suggested that conflict was inherently bad and indicated a failure within the organization, meaning it should be avoided at all costs.
- **Behavioral & Interactionist Approach:** Modern theories suggest that conflict is inevitable due to human interaction and can even be necessary for an organization's effectiveness.

Justification: Why It Is Sensible to Have Managed Conflict

Whether conflict is "sensible" depends on its nature:

1. **Functional (Constructive) Conflict:** This type of conflict supports the goals of a group and actually helps improve performance. It encourages creativity, prevents stagnant thinking, and leads to better decision-making.
2. **Dysfunctional (Destructive) Conflict:** This type of conflict hinders group performance and should be resolved or avoided.

Conclusion: It is not sensible to avoid all conflict, as avoiding it entirely can lead to a lack of innovation and poor-quality decisions. Instead, conflict should be managed within "manageable limits" to be beneficial for the project's success.

Question 2: Define Public-Private Partnership (PPP). Explain the importance/significance of PPP models and the private sector in development activities for countries like Nepal.

Definition of Public-Private Partnership (PPP)

A Public-Private Partnership is a contractual agreement between a public entity (government) and a private entity for the delivery of infrastructure or services in the public interest. In this model, the public partner focuses on the output (results), while the private partner manages the input and carries a substantial portion of the project risk.

Significance of PPP Models for Nepal

For a developing nation like Nepal, PPPs are critical for several reasons:

- **Addressing Infrastructure Gaps:** Nepal's global competitiveness in infrastructure has historically been low (ranked 131 out of 133 in one report), and government investment alone is not enough to meet the increasing demand.
- **Cost-Effectiveness and Efficiency:** Private sector involvement often enhances cost-effectiveness and brings innovative delivery approaches that the public sector may lack.
- **Risk Sharing:** PPPs ensure a balance between risk and reward by transferring appropriate risks (such as construction or market risks) to the private party.
- **Value for Money:** The model aims to demonstrate better "Value for Money" than traditional public procurement.
- **Timely Solutions:** It helps generate timely solutions to public problems, such as waste management or transportation infrastructure like the "fast track" road project.

Question 3: Define dispute and conflict. Explain the levels/sources and various methods (dispute resolution process) for resolving them.

Definitions

- **Conflict:** A state of opposition between ideas, interests, or people. It is the inevitable outcome of behavioral interactions.
- **Dispute:** A dispute occurs when one party makes a claim or demand and the other party contradicts or rejects that claim.

Levels and Sources of Conflict

Level of Conflict	Description
Intrapersonal	Occurs within the individual self.
Interpersonal	Occurs between two or more persons due to communication gaps or goal differences.
Intergroup	Occurs between different groups, often over resources or reward systems.
Inter-organizational	Occurs between different organizations, often due to competition.

Sources of Conflict: These include personal differences, goal/role incompatibility, lack of resources, communication gaps, and organizational climate changes.

Dispute Resolution Process (Methods)

1. **Amicable Settlement:** The preferred first step, where parties try to settle the issue through friendly discussion without formal third-party intervention.
2. **Adjudication:** A neutral third party (adjudicator) reviews evidence and arguments to make a decision, typically for disputes up to 100 million NPR.

3. **Dispute Resolution Board/Committee:** A 3-member committee established for larger projects (more than 100 million NPR) to provide recommendations or decisions.
4. **Arbitration:** A formal process outside of court where an arbitrator or a panel makes a binding decision on the dispute. This is common for issues like late payments or unclear instructions.
5. **Litigation:** The final legal resort involves taking the case to a formal court of law.

Question 4: What are the challenges/merits/demerits of globalization you see on the engineering profession?

Merits of Globalization

- **Access to Global Standards:** Nepalese engineers can follow international standards (like IEEE or ISO), making them more competitive in the global job market.
- **Knowledge and Technology Transfer:** It facilitates the easy transfer of assets, information, and collaboration with international partners.
- **Increased Innovation:** Competition on a global scale drives engineers to innovate farther and faster.
- **Economic Growth:** It allows developing countries to improve their standard of living through international trade and investment.

Challenges and Demerits

- **Cultural Degradation:** There is a risk of losing local cultural identity due to the influence of a globalized "modern" culture.
- **Job Insecurity:** While it creates opportunities, it also creates insecurity as local enterprises may struggle to compete with multinational corporations.
- **Homogeneity:** It can lead to an "increasing homogeneity" where local uniqueness is lost in favor of global trends.
- **Ethical Misuse:** Globalization makes technology misuse (like cybercrime or data breaches) a global threat rather than a local one.

Question 5: “Construction is Destruction to Environment.” Do you agree with this statement? Give reasons.

I **partially agree** with this statement, but only if construction is not balanced with environmental safeguards. Construction projects are essential for development but inherently impact the natural world.

Reasons why Construction can be "Destruction"

1. **Pollution:** Construction can create significant pollution that is hazardous to public health.

2. **Habitat Loss:** Large projects, such as dams or highways, can involve clearing forests and displacing endangered species.
3. **Resource Depletion:** It involves the massive use of natural materials like stone, timber, and water.

How Engineering Mitigates This

To prevent "destruction," modern engineering practice mandates:

- **Initial Environmental Examination (IEE):** A study to see if a project will have significant adverse impacts.
- **Environmental Impact Assessment (EIA):** A detailed study required for large projects (like national highways) to identify and mitigate environmental damage.
- **Sustainable Development:** The goal is to meet current needs without compromising the ability of future generations to meet theirs.

Question 6: Explain globalization and cross-cultural issues.

Globalization

Globalization is the process of interaction and integration among people, companies, and governments of different nations, driven by international trade, investment, and aided by information technology. It has transformed economic life by allowing for easier collaboration and trade.

Cross-Cultural Issues

Culture consists of the ideas, customs, and social behaviors of a particular group. Cross-cultural interaction occurs when people from different backgrounds combine.

- **Communication Gaps:** Different cultures have different norms regarding personal space, gestures, manners, and time.
- **Values and Morals:** A person's upbringing shapes their mindset. In business, failing to understand these norms can lead to misunderstandings.
- **Need for Competence:** Cross-cultural competence is essential for engineers today, as successful international trade and projects depend on the smooth interaction of diverse teams.

Question 7: Why is the participatory approach necessary in development activities? What are their benefits in the context of Nepal?

Necessity of Participatory Approach

A participatory approach involves the active involvement of the local community or "beneficiaries" in the planning and execution of projects. In Nepal, this is often done through **Users' Committees** or **Beneficiary Groups**.

Benefits in the Context of Nepal

1. **Community Ownership:** When local people participate, they feel a sense of relationship and commitment to the project, ensuring its long-term maintenance.
2. **Increased Capacity:** It provides a platform for increasing the people's capacity and skills through hands-on involvement.
3. **Efficiency and Resource Use:** It allows for the use of local materials and labor, which is often more economical for smaller rural projects.
4. **Social Stability:** By involving diverse groups, it helps address the "interdependence" needed for a society to function and survive.
5. **Alignment with Local Needs:** It ensures that development projects actually address the specific problems of the rural or urban society they are meant to serve.